

1.8 The Squeeze Theorem

Calculus

Solutions Practice

Evaluate each limit.

1. $\lim_{x \rightarrow 0} x \cos\left(\frac{1}{x}\right)$

$$-1 \leq \cos\left(\frac{1}{x}\right) \leq 1$$

$$-x \leq x \cos\left(\frac{1}{x}\right) \leq x$$

$$0 \leq \lim_{x \rightarrow 0} x \cos\left(\frac{1}{x}\right) \leq 0$$

Therefore

$$\lim_{x \rightarrow 0} x \cos\left(\frac{1}{x}\right) = 0$$

2. $\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right)$

$$-1 \leq \sin\left(\frac{1}{x}\right) \leq 1$$

$$-x^2 \leq x^2 \sin\left(\frac{1}{x}\right) \leq x^2$$

$$0 \leq \lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) \leq 0$$

Therefore

$$\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = 0$$

3. $\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x^2}\right)$

$$-1 \leq \sin\left(\frac{1}{x^2}\right) \leq 1$$

$$-x \leq x \sin\left(\frac{1}{x^2}\right) \leq x$$

$$0 \leq \lim_{x \rightarrow 0} x \sin\left(\frac{1}{x^2}\right) \leq 0$$

Therefore

$$\lim_{x \rightarrow 0} x \sin\left(\frac{1}{x^2}\right) = 0$$

4. Let g and h be the functions defined by $g(x) = x^2 - 3x$ and $h(x) = 2 - 2x$.

If f is a function that satisfies $g(x) \leq f(x) \leq h(x)$ for all x , what is $\lim_{x \rightarrow 2} f(x)$?

$$2^2 - 3(2) \leq \lim_{x \rightarrow 2} f(x) \leq 2 - 2(2)$$

$$-2 \leq \lim_{x \rightarrow 2} f(x) \leq -2$$

$$\lim_{x \rightarrow 2} f(x) = -2$$

5. Let g and h be the functions defined by $g(x) = \cos(\pi(x+2)) - 3$ and $h(x) = \frac{x^2}{2} + x - \frac{7}{2}$.

If f is a function that satisfies $g(x) \leq f(x) \leq h(x)$ for $-2 \leq x \leq 0$, what is $\lim_{x \rightarrow -1} f(x)$?

$$(\cos(\pi(-1+2)) - 3) \leq \lim_{x \rightarrow -1} f(x) \leq \frac{(-1)^2}{2} + (-1) - \frac{7}{2}$$

$$(\cos(\pi) - 3) \leq \lim_{x \rightarrow -1} f(x) \leq -\frac{6}{2} - 1$$

$$-4 \leq \lim_{x \rightarrow -1} f(x) \leq -4$$

$$\lim_{x \rightarrow -1} f(x) = -4$$

6. Let g and h be the functions defined by $g(x) = x^2 + x - 1$ and $h(x) = -x^2 - 4x - 2$.

If f is a function that satisfies $g(x) \leq f(x) \leq h(x)$ for all x , what is $\lim_{x \rightarrow -1} f(x)$?

$$(-1)^2 + (-1) - 1 \leq \lim_{x \rightarrow -1} f(x) \leq -(-1)^2 - 4(-1) - 2$$

$$1 - 2 \leq \lim_{x \rightarrow -1} f(x) \leq -1 + 4 - 2$$

$$-1 \leq \lim_{x \rightarrow -1} f(x) \leq 1$$

Limit cannot be determined from the Squeeze Theorem.

7. Let g and h be the functions defined by $g(x) = -x^2 - 2x + 5$ and $h(x) = 2x^2 - x - 4$.

If f is a function that satisfies $g(x) \leq f(x) \leq h(x)$ for all x , what is $\lim_{x \rightarrow -2} f(x)$?

$$-(-2)^2 - 2(-2) + 5 \leq \lim_{x \rightarrow -2} f(x) \leq 2(-2)^2 - (-2) - 4$$

$$-4 + 4 + 5 \leq \lim_{x \rightarrow -2} f(x) \leq 8 + 2 - 4$$

$$5 \leq \lim_{x \rightarrow -2} f(x) \leq 6$$

Limit cannot be determined from the Squeeze Theorem.

8. Let g and h be the functions defined by $g(x) = \sin\left(\frac{\pi}{2}(x+1)\right) - 1$ and $h(x) = \cos(\pi x) - 3$.

If f is a function that satisfies $g(x) \leq f(x) \leq h(x)$ for all x , what is $\lim_{x \rightarrow 2} f(x)$?

$$\sin\left(\frac{\pi}{2}(3)\right) - 1 \leq \lim_{x \rightarrow 2} f(x) \leq \cos(2\pi) - 3$$

$$1 - 1 \leq \lim_{x \rightarrow 2} f(x) \leq 1 - 3$$

$$-2 \leq \lim_{x \rightarrow 2} f(x) \leq -2$$

$$\lim_{x \rightarrow 2} f(x) = -2$$

9. Let g and h be the functions defined by $g(x) = x^2$ and $h(x) = \cos(x)$.

If f is a function that satisfies $g(x) \leq f(x) \leq h(x)$ for $-0.4 \leq x \leq 0.4$, what is $\lim_{x \rightarrow 0} f(x)$?

$$0^2 \leq \lim_{x \rightarrow 0} f(x) \leq \cos(0)$$

$$0 \leq \lim_{x \rightarrow 0} f(x) \leq 1$$

Limit cannot be determined from the Squeeze Theorem.

10. Let f and g be the functions defined by $f(x) = \frac{\sin x}{5x}$ and $g(x) = x^2 \cos\left(\frac{1}{x^3}\right)$ for $x \neq 0$. The following inequalities are true for $x \neq 0$. State whether each inequality can be used with the squeeze theorem to find the limit of the function as x approaches 0?

a. $5(\sin(\pi(x+1))) \leq f(x) \leq \frac{1}{5}$

$$5(\sin(\pi)) \leq \lim_{x \rightarrow 0} f(x) \leq \frac{1}{5}$$

$$0 \leq \lim_{x \rightarrow 0} f(x) \leq \frac{1}{5}$$

No

b. $-x^2 \leq g(x) \leq x^2$

$$0 \leq \lim_{x \rightarrow 0} g(x) \leq 0$$

Yes

11. Let f and g be the functions defined by $f(x) = \frac{\cos x - 1}{x^2}$ and $g(x) = x^2 \sin\left(\frac{1}{x}\right)$ for $x \neq 0$. The following inequalities are true for $x \neq 0$. State whether each inequality can be used with the squeeze theorem to find the limit of the function as x approaches 0?

a. $\frac{1}{5} \leq f(x) \leq \frac{1}{2}$

No

b. $-x^2 - \frac{1}{2} \leq f(x) \leq x^2 - \frac{1}{2}$

$$-\frac{1}{2} \leq \lim_{x \rightarrow 0} f(x) \leq -\frac{1}{2}$$

Yes

c. $-x^2 \leq g(x) \leq x^2$

$$0 \leq \lim_{x \rightarrow 0} g(x) \leq 0$$

Yes

d. $-\frac{1}{x} \leq g(x) \leq \frac{1}{x}$

No. $\frac{1}{x}$ is unbounded
as $x \rightarrow 0$.

12. Let f and g be the functions defined by $f(x) = \frac{x^2 \sin x}{x}$ and $g(x) = x \cos\left(\frac{1}{|x|}\right)$ for $x \neq 0$. The following inequalities are true for $x \neq 0$. State whether each inequality can be used with the squeeze theorem to find the limit of the function as x approaches 0?

a. $-x \leq f(x) \leq x$

$$0 \leq \lim_{x \rightarrow 0} f(x) \leq 0$$

Yes

b. $-\frac{1}{x} \leq g(x) \leq \frac{1}{x}$

No. $\frac{1}{x}$ is unbounded
as $x \rightarrow 0$.

c. $2^{-x} - 1 \leq g(x) \leq 2^x - 1$

$$2^0 - 1 \leq \lim_{x \rightarrow 0} g(x) \leq 2^0 - 1$$

$$0 \leq \lim_{x \rightarrow 0} g(x) \leq 0$$

Yes